

PROBLEM SET / 01

Evidence-bounded organizational sensemaking

An auditable agent workflow for organizational sensemaking

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Developed after the original submission and grading dates. Status: In progress.

Abstract

An organization is a collection of related teams, projects, records and external dependencies. A useful assessment must follow those relationships without turning every connected fact into a claim about the entire organization. This implementation equips an agent with six evidence tools, a directed graph, structured retrieval and direct media inspection. The agent chooses its investigative path and produces cited findings, qualifications and unresolved conflicts. [1]

A recorded local-model investigation of the fictional Riverwatch project retrieved nine sources through 15 MCP calls and inspected an image plus sampled video frames. A separate AI source review accepted the bounded development case. The case demonstrates traceable tool use and careful treatment of competing claims; it is neither independent human validation nor a statistical estimate of reliability. [2] [3] [4]

Problem and design objective

The SCADS task concerns sensemaking for nonatomic entities: entities whose properties depend on connected parts. Evidence is distributed across representations, names can be ambiguous, and dates can disagree. The design objective is an inspectable path from an analyst question to source-supported conclusions, with the subject and evidential scope preserved at every step. [1]

Implemented contribution

The working contribution combines identity resolution, bounded relationship traversal, source retrieval, multimodal evidence access and a recorded synthesis workflow. A local llama.cpp model supplies the free execution path; an optional Codex harness uses the same MCP interface. A static case explorer exposes saved reports, sources and tool sequences for inspection without running inference. [1]

Evidence class: authored synthetic investigation, recorded local inference, separate AI source review and deterministic software checks. The paper reports the implemented method and its observed behavior.

1. Research framing and formal scope

ReAct interleaves language-model decisions with actions that gather information from an external environment. That is a useful architectural precedent for this project's model-selected tools; the present work concentrates on typed organizational paths and provenance rather than reproducing ReAct's benchmark claims. [8]

Liu et al. found that a model's use of evidence can depend on where relevant information appears in a long context. This motivates testing the present system's selective retrieval and compact synthesis, but does not prove that its policies improve accuracy. W3C PROV-DM supplies a vocabulary for entities, activities, derivation and attribution; the project ledger is a concrete traceability mechanism, not a claim of complete PROV conformance. [9] [10]

Problem formulation

Let $G=(V,E)$ be a directed typed entity graph; let D be versioned evidence items, each with an ID, source locator, date and content fingerprint. An investigation receives target v , question q and resource budget B . A tool trace T selects a reachable scope V_T , retrieves D_T , and produces claims C with citation map $P:C \rightarrow \text{subsets}(D_T)$. This notation formalizes the implemented contract; it is not an additional trained model. [1] [11]

For each reported claim c :
 $P(c)$ is nonempty and refers to retrieved source versions.
 $\text{subject}(c)$ is within the declared investigative scope.
 cited paths retain relation types and edge evidence.
 unsupported factual resolution is replaced by qualification.

A valid citation map is necessary but insufficient: a source can be read correctly yet fail to entail the claim. The central research question is therefore whether the workflow improves source-supported analytic coverage while keeping attribution, uncertainty and resource cost inspectable. The evidence in this paper addresses feasibility for one known synthetic case.

Prospective hypothesis: compared with an evidence-packet baseline, selective tool use can recover dependency-path evidence with fewer irrelevant source reads at comparable claim support. This is a testable hypothesis, not an achieved result.

2. Method and implemented architecture

The architecture separates decisions made by the model from evidence supplied by the host. The model selects tools and writes findings. Retrieval returns identified records and graph proofs; declared host fields provide the overview and run limitations. This distinction lets a reader inspect which statements were generated and which were assembled from verified run facts. [1] [2]

Stage	Operation	Preserved evidence
Resolve	entity_search	Canonical IDs and ambiguous alternatives
Connect	traverse_relationships	Typed directed paths, depth and supporting sources
Read	search_evidence + read_evidence	Full records, dates and locators
Inspect	inspect_media + read_media	Availability, original pixels and decoded frame times
Synthesize	Recorded agent run	Cited findings, conflicts, qualifications and trace

Structured records are queried through SQLite; source text and supplied assertion triples are indexed locally. The graph is a directed JSON edge-list representation. Relationship expansion has a declared depth, and every relevant edge keeps its supporting source IDs. A dependency edge is distinct from organizational containment or managerial control. [1] [12]

Retrieve before citing

The bounded catalog exposes metadata pages for selective discovery. Discovering an evidence ID is not equivalent to reading the source. Inventory coverage and fully retrieved, citable evidence therefore have separate records. This reduces the chance that an agent cites a promising title as though it had inspected the underlying statement. [7]

Reconcile rather than overwrite

Competing assertions retain subject, date, predicate, value and provenance. A more recent record can suggest an update, but recency alone does not prove that an earlier assertion was corrected. Reports preserve unresolved differences, distinguish plans from observed events and label inferences. Repeated fixture content across media is shared-origin evidence, not independent corroboration. [1] [2]

Figure 1. The table follows the implemented investigative sequence. It is a method diagram, not a claim that every run visits all stages or that the model follows a fixed script.

3. Evidence control and algorithmic choices

The abstract procedure below expresses the host/model boundary used by the prototype. Actions and retrieved observations form the public audit record. Private model reasoning is neither required as proof nor substituted for source evidence. The procedure is descriptive pseudocode; the frozen implementation and schema remain authoritative for reproduction. [5] [11]

```

resolve target; if ambiguous, request clarification
initialize bounded graph scope and retrieved-source ledger
repeat while action and time budgets permit:
  model selects a permitted tool action
  host validates arguments, executes and records the result
  add exact returned sources and edge proofs to the ledger
  inspect available media when the run policy requires it
  accept finish only after required evidence obligations
  synthesize from admitted evidence; validate citations
  retain report, trace, media observations and run metadata

```

Why graph direction matters

A project depending on a sensor batch and a supplier providing that batch are distinct directed relations. Reversing the direction or treating them as containment changes the analytic meaning. A bounded search must retain the path explanation, not merely the set of connected names. This also allows exclusion of a name-matching arts event without erasing genuinely relevant external dependencies. [1] [2]

Context is a resource and an experimental variable

Retrieving every source can increase context cost and duplicate evidence. The accepted run's compact synthesis retained nine exact record versions while removing seven identical repeated occurrences; its review verified the input lineage. A compact representation may improve resource use but could also omit context needed for interpretation. Any causal claim needs an ablation holding model, question, sources and decoding fixed. [2]

The implemented breadth-first traversal scans the full edge list for each expanded node, giving an $O(|V_T| \text{ times } |E|)$ upper bound with hop-bounded path copying, excluding source and model work. An adjacency index is a prospective optimization, not the current cost model. End-to-end work additionally includes source bytes, media decoding and inference; report those resources separately rather than extrapolating one CPU run. [12]

4. Worked example

The fictional Cedar Basin Research Institute contains divisions, teams and projects. Riverwatch depends on the Silt Sensor Batch supplied by Northline Instruments. A register records readiness, later material records delay, and supplier records contain different expected-arrival dates. The task is to explain readiness and dependencies while keeping the conclusion scoped to Riverwatch. [1] [2]

Evidence pattern	Supported interpretation
September 3 ready register; September 5 delayed material	Retain the dated readiness disagreement; do not declare current truth from chronology alone.
Riverwatch -> sensor batch -> Northline	Preserve the dependency and supply relations; the path does not establish organizational control.
Expected arrival September 14; later September 19	These are planning claims. Neither establishes shipment or receipt.
Pending acceptance and required calibration	Describe release conditions without asserting an exclusive cause or proving the sensors defect-free.
PNG and video caption boards	Inspect actual pixels, while retaining their authored synthetic origin.

The accepted report contains four findings and two conflicts. Its review checked the source IDs, values, dated disagreements, supplier path and uncertainty wording. Original report bytes were preserved; a cosmetic color description and imprecise shipping terminology remain disclosed in the review. The analysis still distinguished expected arrival from delivery. [2] [3]

Media and absence

The run inspected the supplied PNG and decoded video frames returned at 0, 15 and 29 seconds. Those are caption fixtures without audio. An inspection-tag annotation had no declared attachment; the tool reported unavailable pixels. Treating that absence explicitly is part of the result, rather than replacing it with a model-generated visual description. [2]

The unrelated arts event mentioning Riverwatch was excluded. The final report did not propagate project-level concern to an institute-wide shutdown or infer supplier misconduct. [2]

5. Recorded results and interpretation

Measure	Observed result	Interpretation
MCP calls / sources	15 / 9	One known synthetic investigation
Agent actions	21	Includes finish checks, observations and synthesis
Elapsed time	2,538.563 seconds	Shared CPU host; not an isolated speed benchmark
Findings / conflicts	4 / 2	Counts do not establish factual reliability
Separate source review	Accepted with caveats	AI review; no human-gold or held-out estimate
Implementation checks	268 tests passed	Software behavior, separate from analytic acceptance

Run counts, timing and review scope are preserved in the original metadata and review. The latest overview records the implementation test total separately. The local case used frozen implementation v12 and a pinned Qwen3.5-9B Q3_K_M configuration; later implementation checks do not retroactively change that execution. [1] [2] [4] [5]

What the evidence supports

The trace establishes that tools were actually called and that their returned records supported an inspectable investigation. The source review establishes that this particular report preserves the crucial distinctions in the authored case: unresolved readiness, a full supplier path, planned dates and nonexclusive explanations. Mechanical citation validation alone would not establish those semantic properties. [2]

Scope of inference

The corpus has ten entities, 16 authored evidence records and seven retrieval regression cases. Its image and video repeat authored information. This makes it useful for controlled behavioral inspection while restricting any inference about natural scenes, audio, source independence or organizational truth. The development case is known and tuned; its acceptance cannot estimate general performance on unseen organizations. [1] [2]

The most useful result is methodological: uncertainty can remain attached to a claim even when the agent produces a concise answer. A readable answer and a detailed audit trail serve different purposes, and the implementation retains both. This interpretation follows the demonstrated case rather than claiming measured analyst-time savings.

6. Proposed evaluation protocol

PROPOSED STUDY. The protocol below defines future research measurements. None of its hypotheses, comparisons or numerical acceptance decisions are reported as completed experiments.

Research question	Controlled comparison	Primary observation
Does tool use improve coverage?	Same question/model: static packet versus adaptive retrieval	Supported target claims and necessary path coverage
Does compaction preserve meaning?	Exact evidence histories with compaction on/off	Support, contradiction handling and omitted qualifications
Does media add unique value?	Text-only versus actual pixels, using nonduplicate evidence	Evidence-dependent claim changes and source entailment
Does ambiguity handling transfer?	Held-out aliases and decoy entities	False resolution rate and useful abstention

Freeze organization families, source revisions and analyst questions before the final evaluation. Split by organization rather than individual document so that repeated templates or names do not cross the boundary. Use independent reviewers to annotate required claims, relevant paths and source-supported interpretations, with adjudication for disagreements. Retain source truth separately from agreement among copied documents.

Define supported-claim precision as adjudicated supported factual claims divided by all evaluated factual claims; define coverage over the independently enumerated relevant claims. Score citations for both exact source resolution and entailment. Report ambiguous or unverifiable claims separately. For repeated runs, cluster uncertainty estimates by question and organization, not by correlated claims from one answer.

A paired analyst study should use matched tasks and counterbalanced order, measuring active analyst time, source-opening actions and answer quality under the same information access. A speed improvement without equivalent support quality would not establish reduced analytic burden. Resource accounting must include setup, failures and retries. Predeclare a practically meaningful effect and feasible sample size before collecting final outcomes.

7. Research roadmap and validity

Phase	Concrete output	Decision rule
Corpus qualification	Versioned public-source organizations, independent question/claim labels	Freeze families and provenance before final testing.
Bounded execution	Measured context/tool/media budgets and resumable evidence storage	Admit runs only under an explicit resource profile.
Controlled comparison	Packet/adaptive and compaction ablations with retained failures	Select the simplest approach supported by paired evidence.
Analyst validation	Counterbalanced task study and reproducible source packages	Claim utility only within measured quality/time scope.

This roadmap extends an implemented prototype. It does not relabel the current software as unstarted or predict a completion date. Each phase produces an inspectable artifact before a stronger claim becomes appropriate. New models or larger contexts should be frozen as distinct experimental conditions rather than silently replacing an unsuccessful run.

Threats to validity

Construct validity is the largest current constraint: synthetic captions and authored assertions simplify the relation between text and world truth. Internal validity is limited by tuning on the same case and concurrent host workload. External validity depends on organization diversity, realistic ambiguity and source quality. Reviewer independence matters because AI review and model synthesis may share error patterns.

The source ledger supports reproducibility of what was supplied to a model, not deterministic reproduction of every model output. Runtime version, weight identity, launch profile, sampling and source hashes should accompany each trial. A later better result does not erase earlier evidence; both are part of the research record.

Conclusion

The prototype demonstrates an auditable organizational investigation with actual graph, record and media tools. Its strongest result is preserving claim scope and unresolved evidence through a complete recorded workflow. Establishing broader analytic value requires the controlled evaluations above; the current contribution is the implemented method and its inspectable development evidence.

8. Reproducibility and research record

Start at the project directory in the public source repository. The README provides dependencies, local-model setup and the static case explorer. To inspect the exact analytic result, begin with the original report, trace, ledger and run metadata; use the frozen v12 instructions when reproducing the accepted model configuration. [\[1\]](#) [\[2\]](#) [\[5\]](#)

```
python sensemaking.py entities
python sensemaking.py investigate riverwatch --max-hops 2
python sensemaking.py evaluate
python scripts/build_showcase.py
```

The deterministic investigate command creates an evidence packet; it is not the end-to-end agent analysis. Media inspection requires the documented ffmpeg/ffprobe tools. The local inference path requires separately installed model assets. Source-package reproduction checks bind file hashes and viewer contents without treating a same-host extraction as independent model validation. [\[1\]](#) [\[6\]](#)

Citation links point to the nested project source. The publication evidence manifest records hashes of the source files used for this paper; editable prose and the PDF builder accompany the publication.

A reproducible research record

The public nested project source contains implementation, documented contracts and retained development evidence. The publication evidence manifest records SHA-256 fingerprints for each local source cited in this paper. External references link to primary publications or official technical documentation, verified for this edition.

A future rerun should record the exact commit, dependency/runtime identities, input hashes, configuration and environment, plus each attempt and its outcome. Keep generated outputs separate from source and mark any changed implementation as a new condition. Reproduction of a software check does not automatically reproduce a model judgment, benchmark distribution or independent human label.

Authorship and project status

Boomer Rawlings developed this work individually after the original submission and grading dates. The current project is In progress. The implemented methods and recorded observations in this paper are separated from the explicitly proposed research protocol and roadmap. No group-project, graded-submission or completed-research claim is made.

References

Repository references identify the evidence used for the implementation and result claims. Primary literature supports the research framing; its published results are not attributed to this project.

[1] [Project overview and implemented system](#)

IMPLEMENTATION.md

[2] [Riverwatch14 independent AI source review](#)

examples/local-riverwatch-guidance-accepted/REVIEW.md

[3] [Original Riverwatch14 report](#)

examples/local-riverwatch-guidance-accepted/report.md

[4] [Original run metadata](#)

examples/local-riverwatch-guidance-accepted/run-metadata.json

[5] [Frozen v12 reproduction instructions](#)

evaluations/local-pilot/implementation-v12/README.md

[6] [Source-package reproduction](#)

docs/package-reproduction.md

[7] [Evidence catalog contract](#)

docs/evidence-catalog.md

[8] Yao et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. ICLR; arXiv:2210.03629.

<https://arxiv.org/abs/2210.03629>

[9] Liu et al. (2024). Lost in the Middle: How Language Models Use Long Contexts. TACL; arXiv:2307.03172.

<https://arxiv.org/abs/2307.03172>

[10] Moreau and Missier, eds. (2013). PROV-DM: The PROV Data Model. W3C Recommendation.

<https://www.w3.org/TR/prov-dm/>

[11] [Project integration and report contracts](#)

docs/integration.md

[12] [Implemented directed traversal and evidence retrieval](#)

sensemaking.py